Analysis Of Global Earthquake Measurements

Mischa Hartmann gdv HS25

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Contents

1 Introduction **2**

2 Dataset **2**
2.1 Features 2
2.2 Description 2

3 Data preparation **2**
3.1 Explorative Data analysis 2
3.1.1 Correlation matrix 2

4 Geological analysis **4**
4.1 Pazific ring of fire 5

5 Design rationale **5**

6 Evaluation results **6**

1 Introduction

Earthquakes are among the most significant natural hazards and are an expression of the dynamic processes within the Earth. They are mainly caused by the build-up and release of stress along tectonic plate boundaries, whereby stored elastic energy is released in the form of seismic waves. The systematic recording and analysis of earthquake measurements therefore forms a central basis for understanding the underlying geophysical mechanisms and for assessing seismic risks.
Modern seismological measurement networks provide high-resolution data on the location, time, magnitude, and depth of earthquake events. These measurements make it possible to identify spatial and temporal patterns of seismic activity and relate them to tectonic structures. In particular, linking earthquake data with different types of plate boundaries—such as spreading zones, subduction zones, and transform faults—provides important insights into the origin and characteristics of different earthquakes.
The aim of this analysis is to systematically evaluate global earthquake measurements and investigate their spatial distribution in relation to tectonic plate boundaries. A particular focus is on distinguishing between earthquakes that trigger tsunamis and those that do not. The combination of statistical analysis and cartographic representation is intended to reveal patterns that contribute to a better understanding of seismic processes and their potential effects.

2 Dataset

2.1 Features

magnitude	Recorded earthquake magnitude (Richter scale)
cdi	Community Decimal Intensity
mmi	Modified Mercalli intensity (instrumental
sig	Event significance level
nst	Number of seismic measurement stations
dmin	Distance to the nearest seismic station
gap	Azimuthal gap between stations (degrees)
depth	Focal depth of the earthquake (km)
latitude	Epicenter latitude
longitude	Epicenter longitude
year	Year of event
month	Month of event
tsunami	Tsunami event (target variable)

2.2 Description

Our dataset contains a total of 782 earthquakes from 2001 to 2022. The proportion of tsunami events is 38.9 % (304 cases), while the proportion of non-tsunami events is 61.1 % (478 cases). The dataset has global coverage and contains no missing values. Here is some additional information I would like to point out regarding the dataset. The recorded magnitude ranges from 6.5 to 9.1 on the Richter scale. The average magnitude is 6.94. Twenty-eight large earthquakes ($\geq$ 8.0) were recorded, including the mega-earthquakes of 2004 (9.1) and 2011 (9.1).

3 Data preparation

3.1 Explorative Data analysis

To uncover hidden patterns in our seismic data, our first step is to locate the most important predictors that influence the outcome of our target variable, tsunami. Since most features are continuous, calculating the Bravais-Pearson correlation is the most efficient method.

3.1.1 Correlation matrix

The diagonal entries should be ignored, as the correlation with itself is always 1. The visualization provides us with a comprehensive summary of valuable information and is particularly useful for indicating a direction in which further visualizations will be meaningful. A further optimization could be to reduce the matrix to the most important elements in order to process less visual data and reduce the risk of information confusion, OR we could split the matrix into separate rows and examine them more closely. Since we have a target variable, this option seems like a good choice, and we can easily sort by the largest absolute correlation.

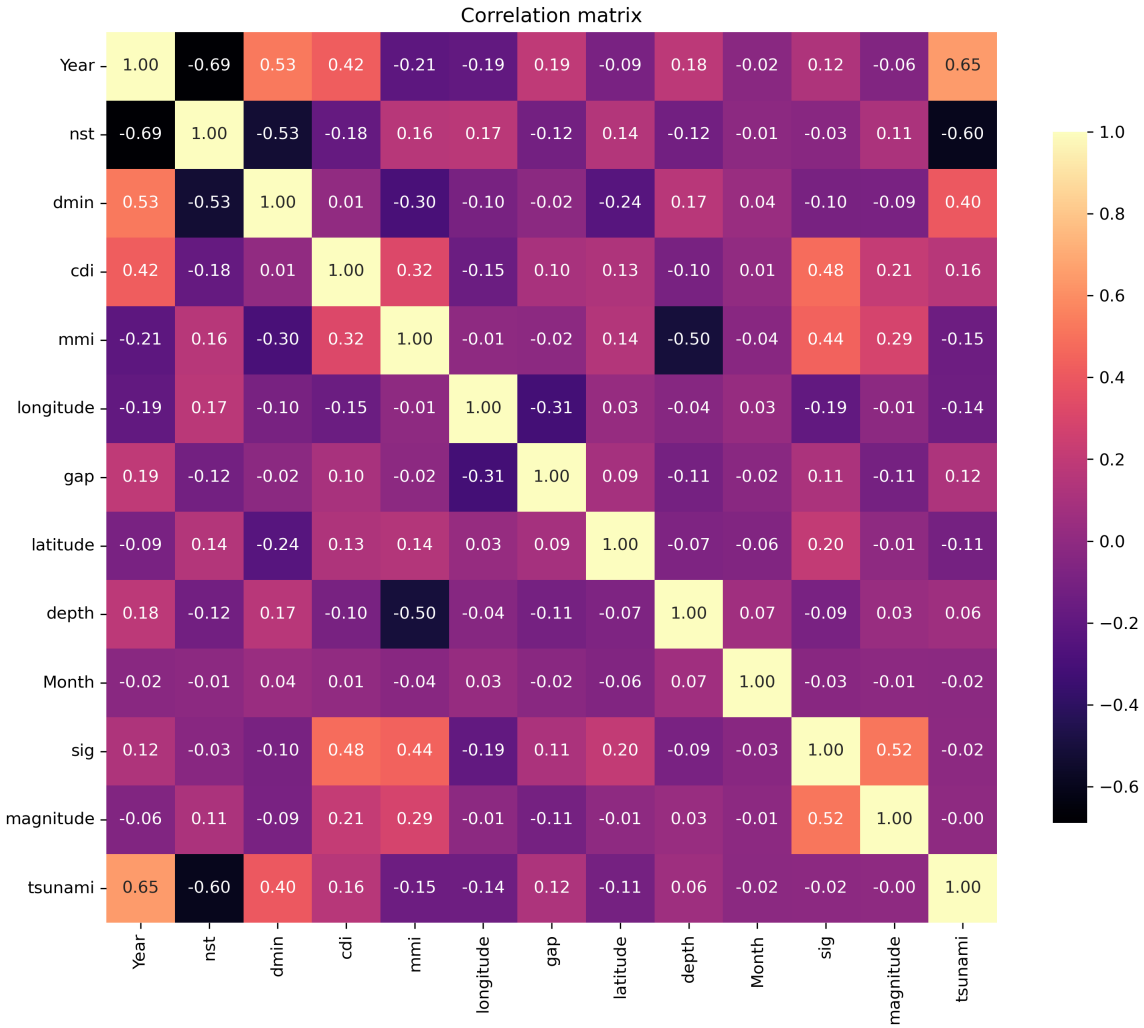


Figure 1: Correlation matrix

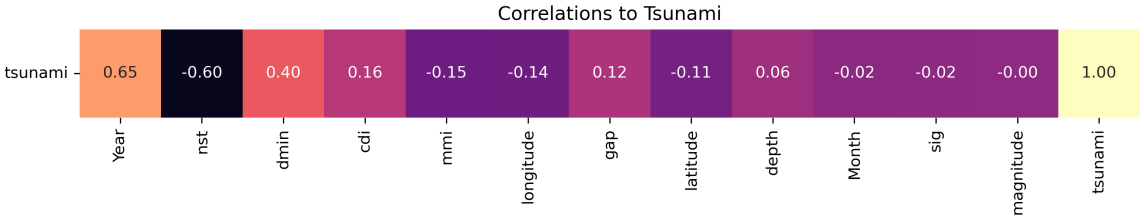


Figure 2: Correlation with tsunami

The visualization shows that the three strongest correlations are the year, the number of seismic stations used to locate the earthquake (nst), and the minimum distance to a seismic station (dmin). With this information, our next task is to determine the relationship between the measured correlation of these three characteristics and our tsunami predictor, as well as the usefulness of this correlation. First of all, the strong positive correlation with the year probably does not reflect a physical relationship. Tsunamis have not suddenly become more frequent in recent decades. Instead, this correlation almost certainly reflects an effect of data collection. Over time, global seismic monitoring networks have been expanded, detection quality has improved, and more tsunami events have been reliably recorded. This means that the year serves as an indicator of technological and reporting improvements rather than a true causal predictor. Its usefulness in a predictive model is therefore statistical rather than geological in nature: it helps explain patterns in the data set but should be interpreted with caution. In contrast, the strong negative correlation with nst has a direct geographical significance. Offshore earthquakes are typically recorded by fewer stations, while onshore events are recorded by many stations. Since tsunamis are usually triggered by offshore or subduction zone earthquakes, low station counts indicate events that are more likely to occur in tsunami-prone areas. This makes nst a truly meaningful predictor, whose correlation reflects both the data geometry and physical reality. Similarly, dmin, the minimum distance from the epicentre to the nearest seismic station, shows a moderate positive correlation with the occurrence of tsunamis. Large values of dmin indicate that the earthquake occurred far away from any station, which in turn points to offshore, remote or deep-sea areas. Since these are precisely the environments where tsunamis most commonly originate, dmin provides useful spatial context. Unlike Year, this feature is more likely to capture a real physical constraint than a temporal bias in reporting. Together, these three characteristics show that correlations can result from both real geophysical patterns and observational biases in the data. For tsunami prediction, nst and dmin provide meaningful location-specific signals, while the year mainly captures changes in reporting quality over time and should therefore be interpreted with caution.

4 Geological analysis

Since our analysis shows that location plays an important role in the occurrence of tsunamis – and that both longitude and latitude have measurable correlations with the target variable – the next logical step is to examine the spatial dimension more closely. By visualising the measurements on a map, we can identify geographical patterns, regional clusters and tectonic relationships that cannot be fully captured by numerical correlations alone. Mapping the data allows us to better understand where tsunami-associated earthquakes tend to occur and how spatial factors interact with the previously identified characteristics.

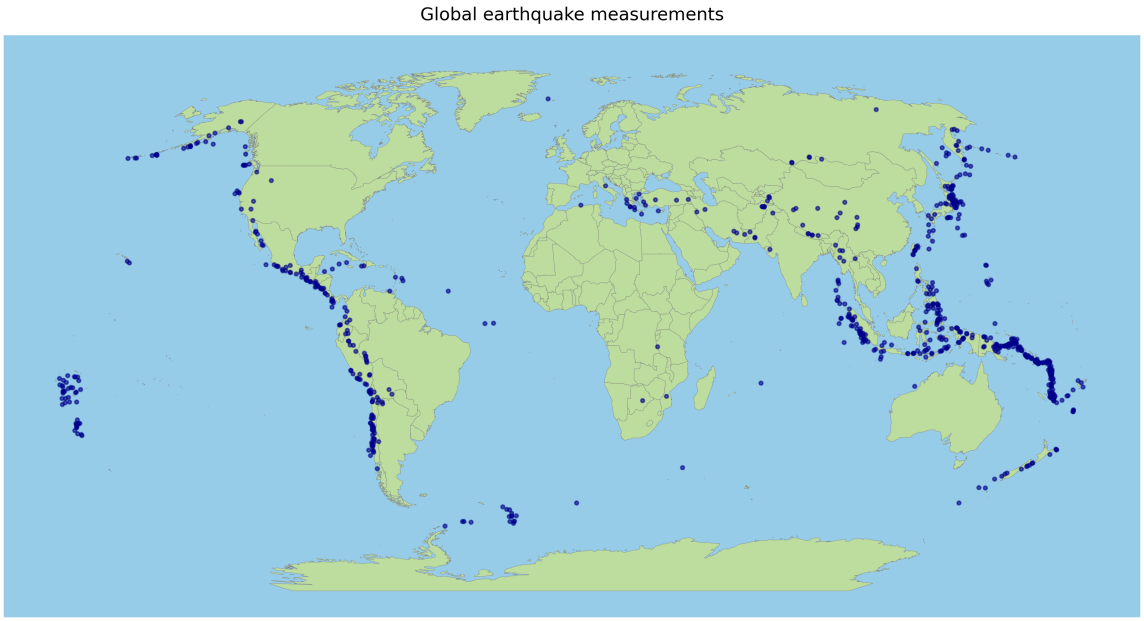


Figure 3: Global earthquake measurements

The argument that many earthquakes appear to follow a similar pattern. By combining this visualisation with knowledge of tectonic plates, we conclude that visualising lines for tectonic plates is useful for further context. This visualisation provides us with a clear overview and divides our target variable into two categories. The quality of the visualisation can be seen in the fact that it draws attention to the increased tsunami distribution along the Pacific coast.

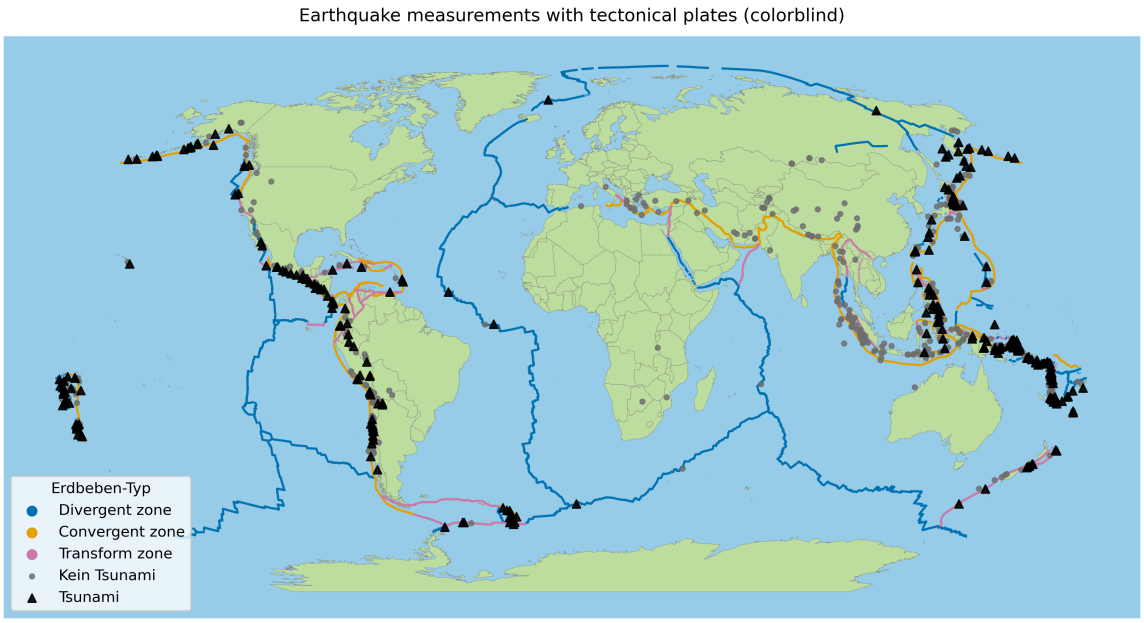


Figure 4: Global earthquake measurements with tectonical plates

4.1 Pazific ring of fire

The Pacific Ring of Fire is a volcanic belt that encircles the Pacific Ocean. It is a direct result of plate tectonics, specifically the movement and changes of lithospheric plates beneath and around the Pacific Ocean. (Geographic, n.d.)

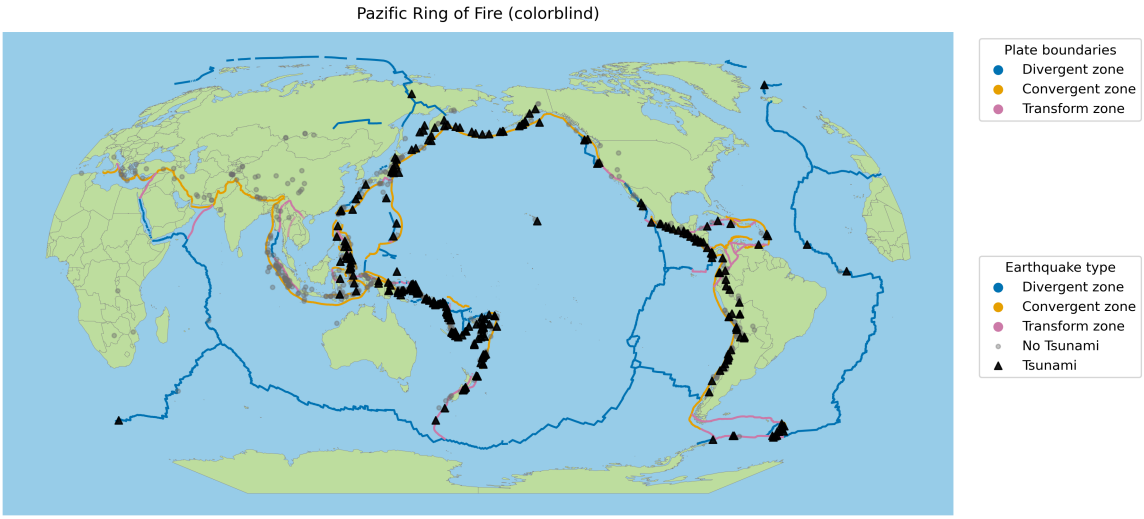


Figure 5: Global earthquake measurements centered in the pacific

5 Design rationale

For the visualisation of the correlation matrix, the magma colour palette from the viridis package was chosen. This palette is perceptually uniform and remains clearly interpretable for viewers with common forms of colour vision deficiency, including Deuteranopia, Protanopia and Tritanopia. In addition, magma preserves meaningful contrast when desaturated or converted to greyscale, which improves robustness across different display conditions and reproduction formats. (Nuñez et al., 2018)
For the cartographic display of the global map, the Robinson projection was selected. This projection provides a balanced compromise between area, shape and distance distortions, resulting in a visually intuitive and aesthetically pleasing representation of the entire world that is well suited for thematic global data. (Streets, 2024)

To further enhance accessibility and clarity, colours were refined using the Okabe–Ito colour palette, which is specifically designed to be colour-blind safe. Tectonic plate boundaries were characterised using this palette to ensure clear differentiation without relying on problematic colour contrasts. The target variable tsunami was deliberately encoded using form and contrast rather than colour alone, employing grey and black tones. This choice maximises contrast, ensures clear distinction from other map elements and maintains interpretability for all viewers, including those with colour vision deficiencies. ("color palette – siegal lab", n.d.)

6 Evaluation results

The evaluation of the visual and spatial analysis reveals a pronounced concentration of tsunami risk along the Pacific coastline. This pattern becomes particularly evident when earthquake locations are examined in conjunction with tectonic plate boundaries. A large proportion of tsunami-generating earthquakes are located offshore along converging plate boundaries, where subduction processes dominate. These zones are characterised by intense seismic activity and vertical displacement of the seafloor, which provides the primary physical mechanism for tsunami generation.

Furthermore, the analysis shows that the majority of recorded tsunamis are associated with convergent plate boundaries rather than divergent or transform boundaries. This observation reinforces the established geological understanding that subduction zones represent the highest tsunami hazard on a global scale. The spatial clustering of tsunami events around the Pacific Ocean strongly aligns with these tectonic conditions.(U.S. Geological Survey, 2025)

The clear visual emergence of this pattern suggests that similar empirical observations in earlier times may have contributed to the historical identification of the Pacific Ring of Fire. Even without modern instrumentation, the repeated occurrence of earthquakes, volcanic activity and tsunamis along the Pacific margins would have indicated a coherent geological system. Thus, the findings of this evaluation not only validate contemporary tectonic theory but also illustrate how systematic observation of natural hazards could have led to the early recognition of the Pacific Ring of Fire as a distinct and highly active geological zone.

Datasets

Global Earthquake-Tsunami Risk Assessment Dataset
Tectonic Plates

References

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List of Figures

1 Correlation matrix 3
2 Correlation with tsunami 3
3 Global earthquake measurements 4
4 Global earthquake measurements with tectonical plates 5
5 Global earthquake measurements centered in the pacific 5